AIDAN COOLEY

Salt Lake City, UT | (505) 695-4465

4idan.cooley@gmail.com | linkedin.com/in/aidan-cooley | https://notcooley.github.io | https://github.com/notcooley

EDUCATION

Bachelor of Science: Computer Science, 3.81 GPA — Salt Lake City, UT

University of Utah, Kahlert School of Computing — *Anticipated December 2026*

Minor in Applied Mathematics

Related Coursework

Algorithms, Computer Systems, Operating Systems, Data Wrangling, Computer Organization, Models of Computation, Real Analysis, Linear Algebra, ODEs

TECHNICAL SKILLS

Programming Languages: C, VHDL, SystemVerilog, Python, C++, Java, R, C#, SQL, QT

Technical Skills: Github, Vivado, Embedded Design, Firmware Design, Computer Engineering and Architecture, Scrum

EXPERIENCE

Los Alamos National Lab — Los Alamos, NM

CAI-3 Intern — *May 2024 – Present*

- Low level computer engineering for space missions
- Design efficient algorithms for satellite modelling and scheduling
- Work on a large team of engineers for products close to launch day
- Work on microcontrollers, FPGAs, and models using C, C++, Python, and VHDL

Wasatch Magazine — Salt Lake City, UT

Editor-in-Chief — *May 2025 – May 2026*

- Run a team of more than a dozen contributors
- Oversee the entire lifecycle of content from pitch to publish.
- Edit all the content written for the magazine
- Work alongside professionals in the industry on advertising, printing, distribution, and collaboration
- Write, take photos, and make videos for our publication

TECHNICAL PROJECTS

Debugging FPGAs Through a Software Slowdown – *Skills used: VHDL, C* — *May 2025 – Aug 2025*

- Develop a test suite for a spacewire router on an FPGA for satellites featuring:
    - Fully customizable data packets, router paths, and speeds
    - Comprehensive error checking through log files
    - User friendly shell allowed for engineers to communicate with the router

Spacewire on the Raspberry Pi Pico – *Skills used: C, microprocessor assembly* — *May 2024 – Aug 2024*

- Implement a system to simulate behavior of spacewire on a $4 microprocessor
- Utilize every piece of microprocessor to run a very complex data transfer protocol
- Showcases lowest bound of needed hardware to implement complex protocol
- Potential to save money through debugging and testing on cheaper devices

Accessing ML Models' Carbon Footprint on an FPGA – *Skills used: HPC, Verilog, Pytorch* — *Sep 2025 – Dec 2025*

- Government funded R&D for mission critical deterrence
- Develop algorithms and data structures to model satellite capabilities
- Implement fast validation schemes to pair with real time scheduling

VOLUNTEER WORK

Utah Climbing Team — Salt Lake City, UT

Outdoor Trip Manager — *August 2023 – May 2026*

- Plan and lead all the outdoor trips for the climbing team
- Teach outdoor climbing technique and safety

BSA Eagle Scout — *Received July 2022*